FEA Mesh Convergence Study

Compact Tension (CT) specimen — plane-strain loading — Abaqus/Standard

Objective: Evaluate h-refinement mesh convergence on a CT fracture-toughness specimen, comparing linear vs. quadratic triangular elements to identify an efficient precision/cost meshing strategy before trusting result accuracy.

Method: Half-symmetry model, frictionless surface-to-surface contact between the loading pin and specimen bore, vertical displacement applied to the pin. Six mesh densities (average element size 1.5 mm down to 0.15 mm) run for both linear and quadratic triangular elements; convergence tracked via relative error between successive refinements on nodal displacement (U2) and local stress (S22).

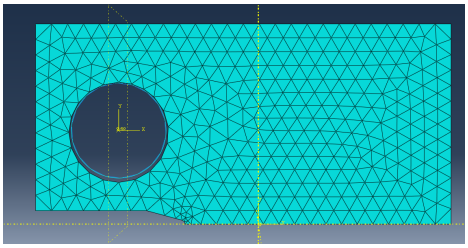


Fig. 1 — Mesh detail around the loaded bore.

Convergence results (selected mesh sizes)

Element	Size (mm)	Nodes	U2 error	S22 error
Linear	1.5	486	—	—
Linear	0.9	892	3.08%	3.19%
Linear	0.15	19,531	0.46%	17.81%
Quadratic	1.5	486	—	—
Quadratic	0.9	2,764	0.39%	2.83%
Quadratic	0.15	76,681	0.29%	6.48%

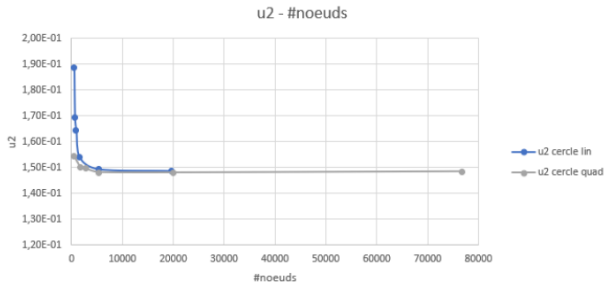


Fig. 2 — U2 convergence: quadratic elements reach <0.5% error at 2,764 nodes; linear elements need 19,531.

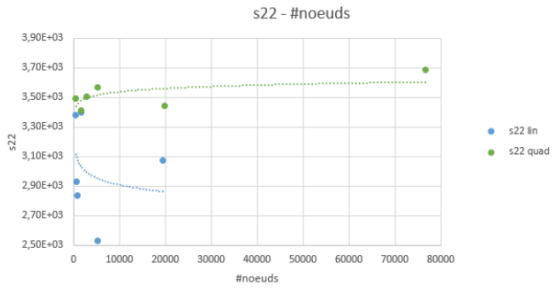


Fig. 3 — S22 stress error vs. node count: no reliable convergence for either element type — signature of a notch-root singularity.

Key finding: Quadratic elements deliver sub-0.5% displacement error with roughly 7× fewer nodes than linear elements — a clear precision/cost win for global refinement. But the local stress metric (S22) never stabilizes for either element type, which is the expected signature of a geometric singularity at the notch root: global mesh density alone cannot be trusted there. Reporting a stress-based result (e.g. a safety factor near a fillet or rib) requires recognizing this pattern and applying targeted local refinement instead of assuming a finer global mesh will fix it.

Collaborative coursework project (2 students) — mesh setup, convergence analysis, and results interpretation shown here. Mechanical Engineering program, ENIM (Metz, France), 2026. Solved in Abaqus/Standard.